



Deployment-Oriented Neural Network State Estimation for Battery Management Systems

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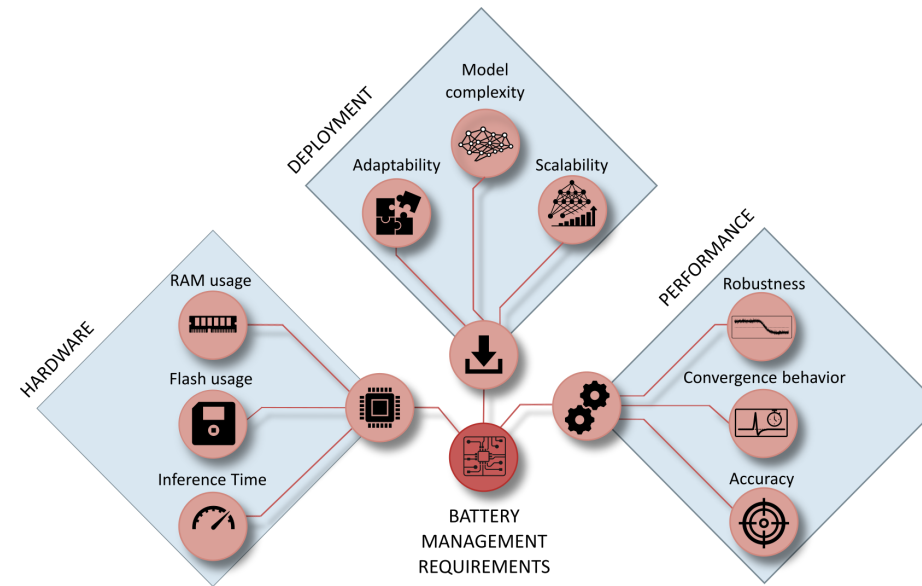
Structure

1. Motivation and research questions
2. Evaluation framework and experimental basis
3. Compact neural ageing estimation
4. Robustness benchmarking
5. Embedded AI and model compression
6. Integrated findings and outlook

Motivation and Research Questions

Why Deployment-Oriented State Estimation?

- SOC and SOH as non-measurable internal battery states
- Field signals with bias, noise, missing data, and timing irregularities
- Embedded limits in flash, RAM, latency, and energy
- Accuracy alone as an incomplete deployment criterion



One estimator path

Data representation to MCU execution

1. Motivation

2. Framework

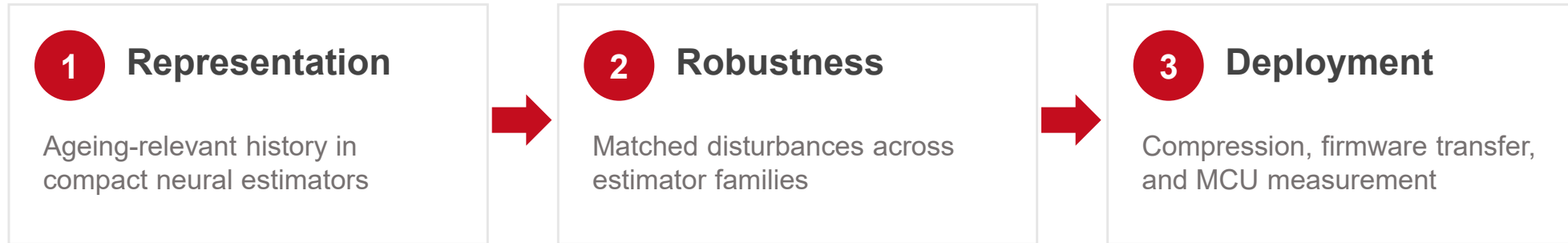
3. Ageing

4. Robustness

5. Embedded AI

6. Synthesis

Research Questions and Contribution Path



- RQ1: Which input representation captures battery ageing?
- RQ2: Which estimator remains reliable under disturbed measurements?
- RQ3: Which compression route preserves accuracy within embedded limits?

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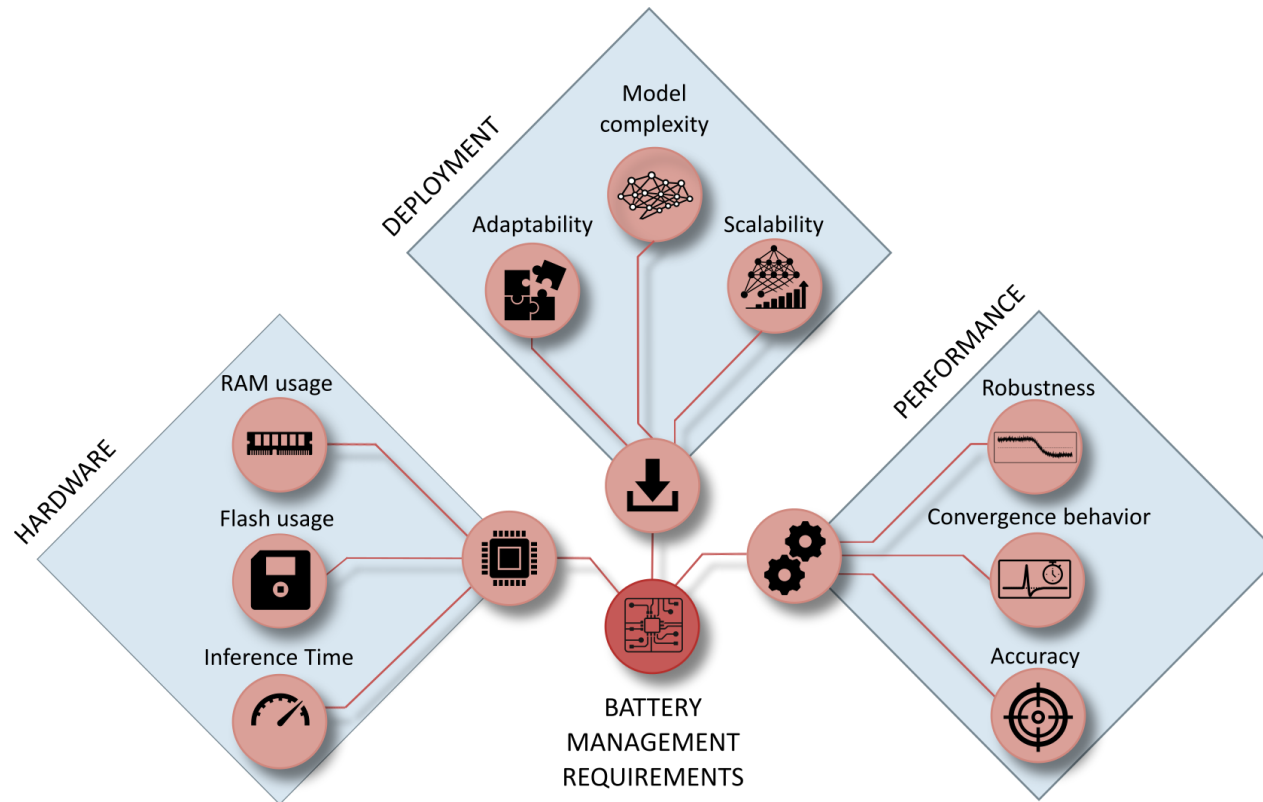
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Evaluation Framework and Experimental Basis

Three Coupled BMS Requirement Dimensions



Joint evaluation

- Performance: accuracy, recovery, disturbance response
- Hardware: flash, RAM, latency, energy
- Deployment: features, export, firmware, repeatability

3 dimensions

No single metric as a substitute

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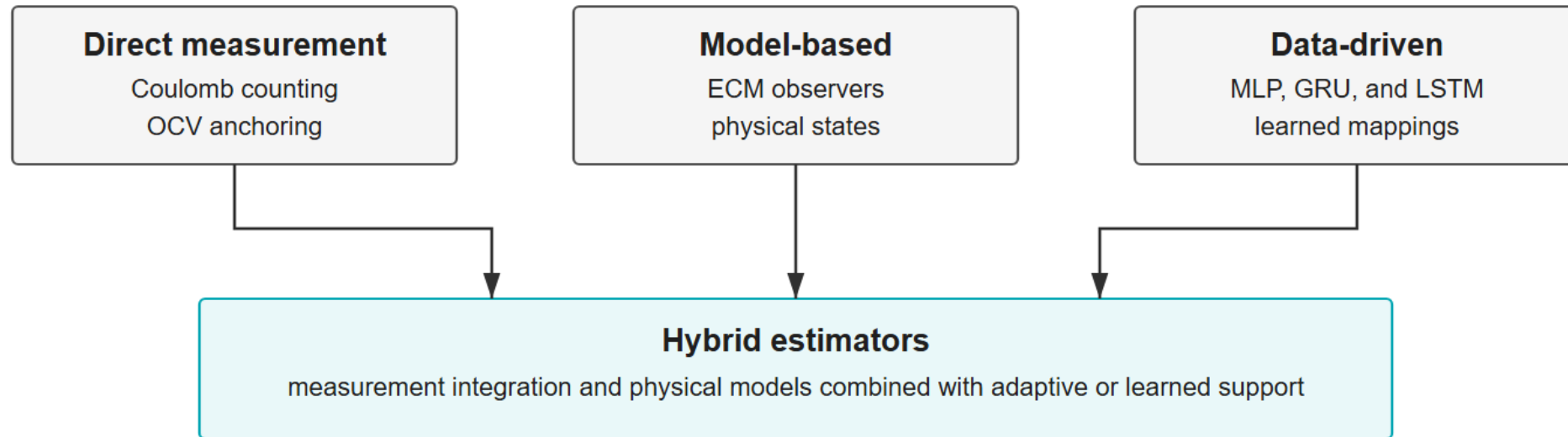
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Estimator Families and Evaluation Logic



- Direct-measurement: low complexity, accumulated drift
- Model-based and hybrid: physical correction paths
- Data-driven: flexible nonlinear mapping and temporal context
- Matched assessment: baseline, disturbance sensitivity, recovery

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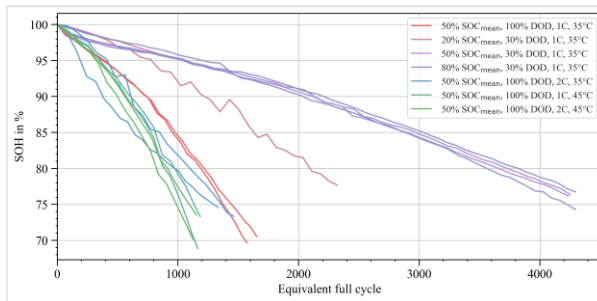
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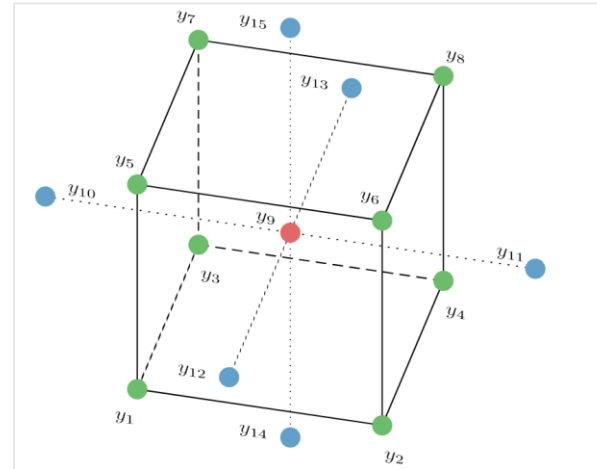
Experimental Basis and Embedded Execution Chain

NMC ageing study



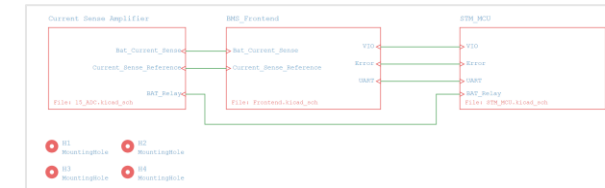
- 14 cells
- 7 ageing scenarios
- Capacity-based SOH labels

LFP DoE campaign



- 15 cells
- 8 operating points
- 1 Hz measurement stream

Embedded chain



- Custom 4s BMS platform
- STM32H7 execution
- Common sensing and firmware basis

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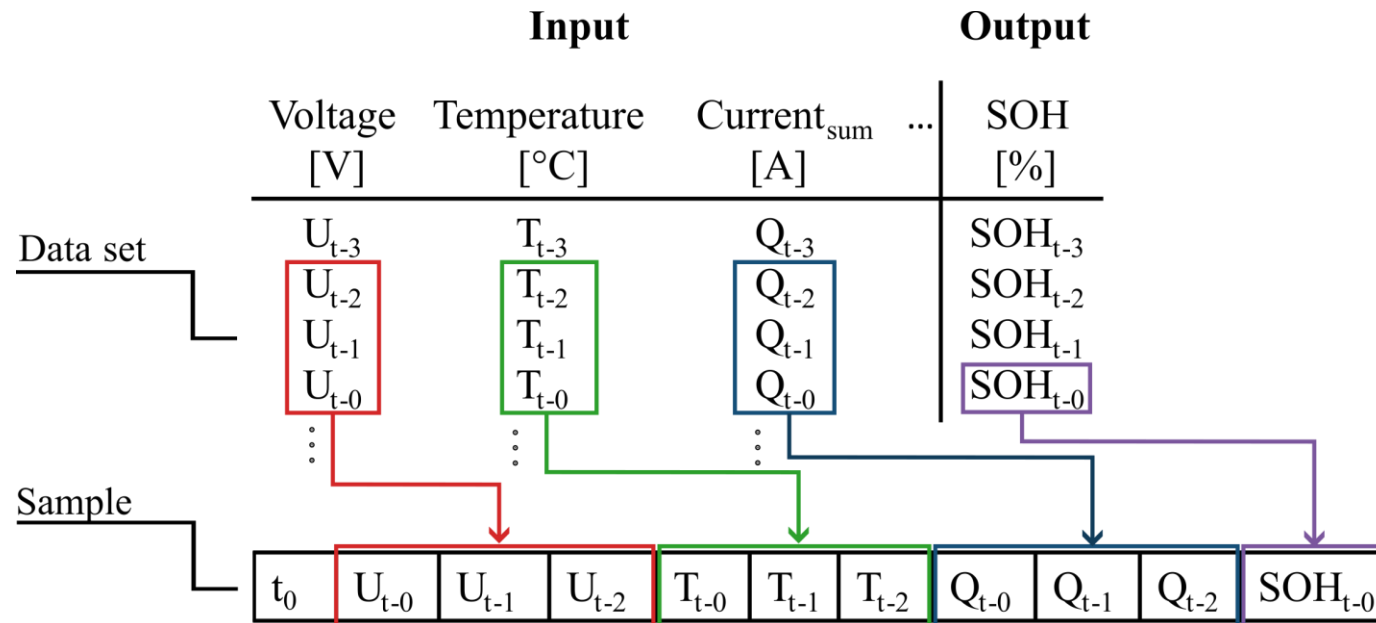
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Compact Neural Ageing Estimation

Time-Aware Features for a Feedforward Model

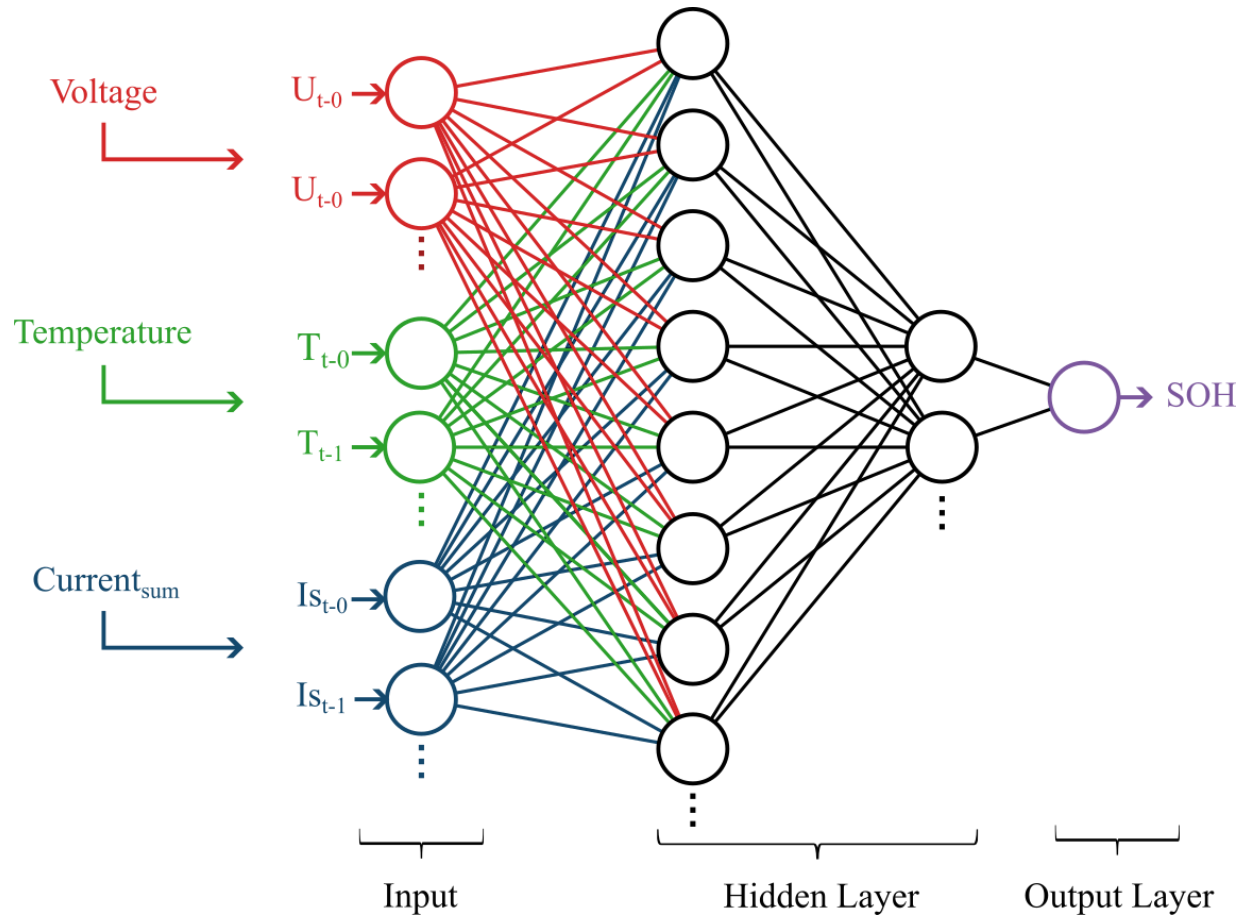


Engineered temporal context

- Voltage, current, and temperature measurements
- Cumulative-current throughput features
- 16 lag steps at 10 min resolution
- 160 min effective history

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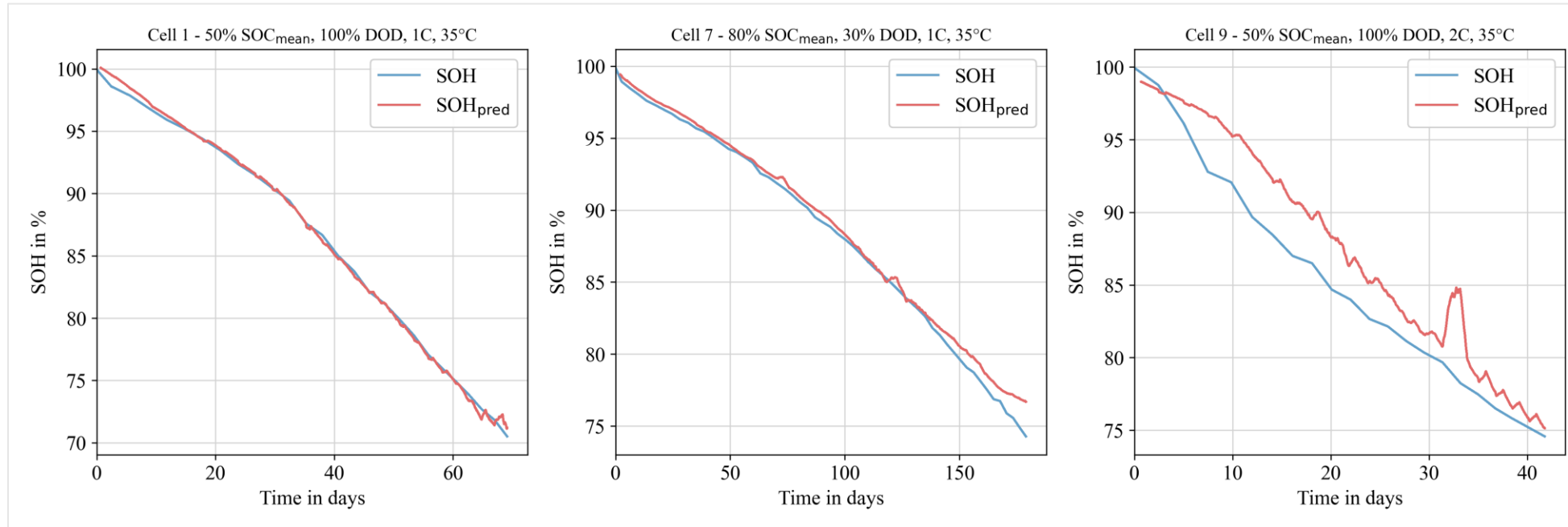
Compact MLP Architecture for SOH Estimation



- Compact feedforward architecture
- Time dependence encoded in engineered inputs
- SOH output from operational measurements
- Low architectural complexity

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Ageing-Estimation Results



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1.10 pp

Average SOH MAE

16 x 10 min

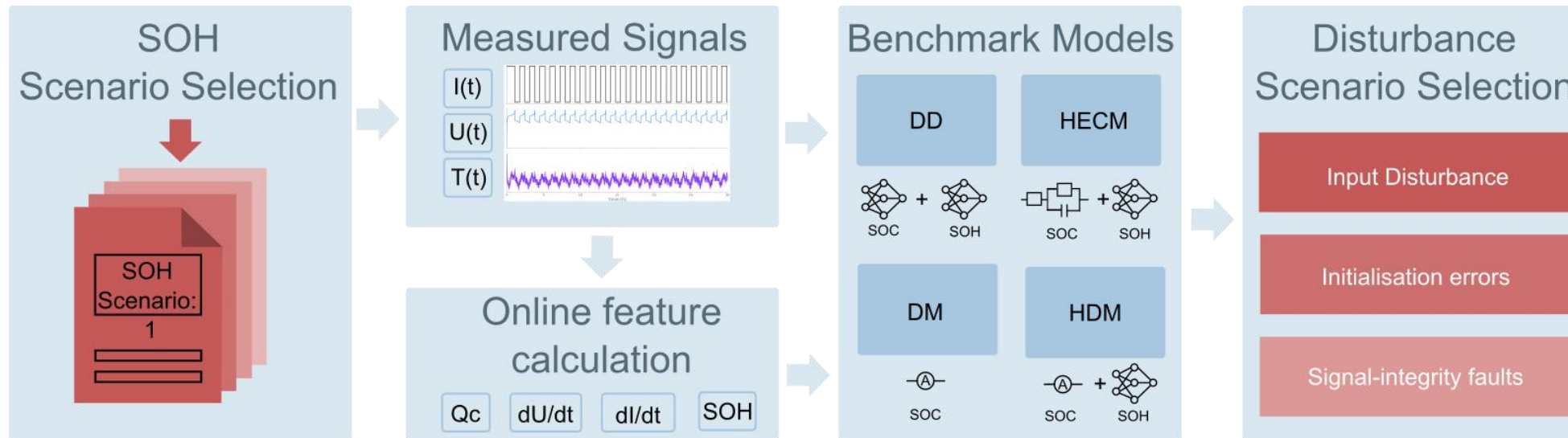
Best lag configuration

Cell 9

Reduced transfer to atypical ageing

Robustness Benchmarking

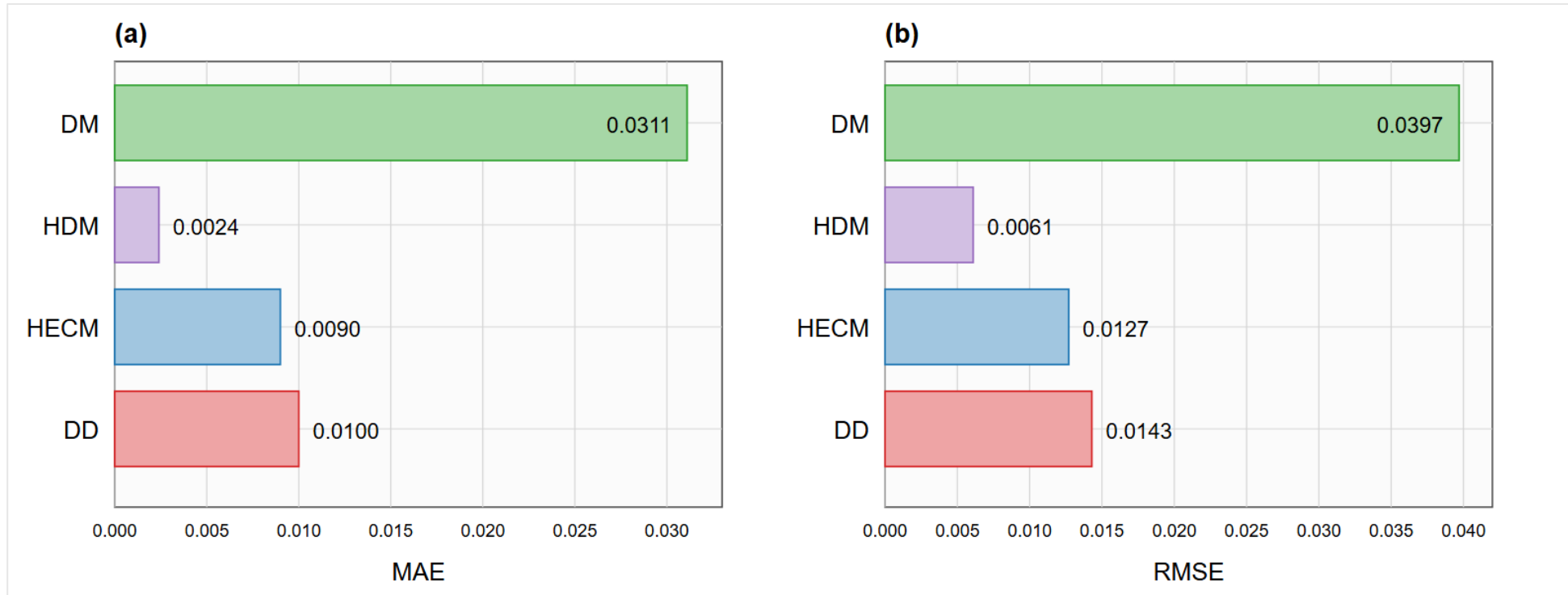
Matched Benchmark Across Four Estimator Classes



- DM: direct Coulomb-counting path
- HDM: direct path with shared SOH context
- HECM: equivalent-circuit observer with SOH context
- DD: recurrent data-driven SOC estimator
- Identical online stream and matched disturbance protocols

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Clean Accuracy Is Only the Starting Point



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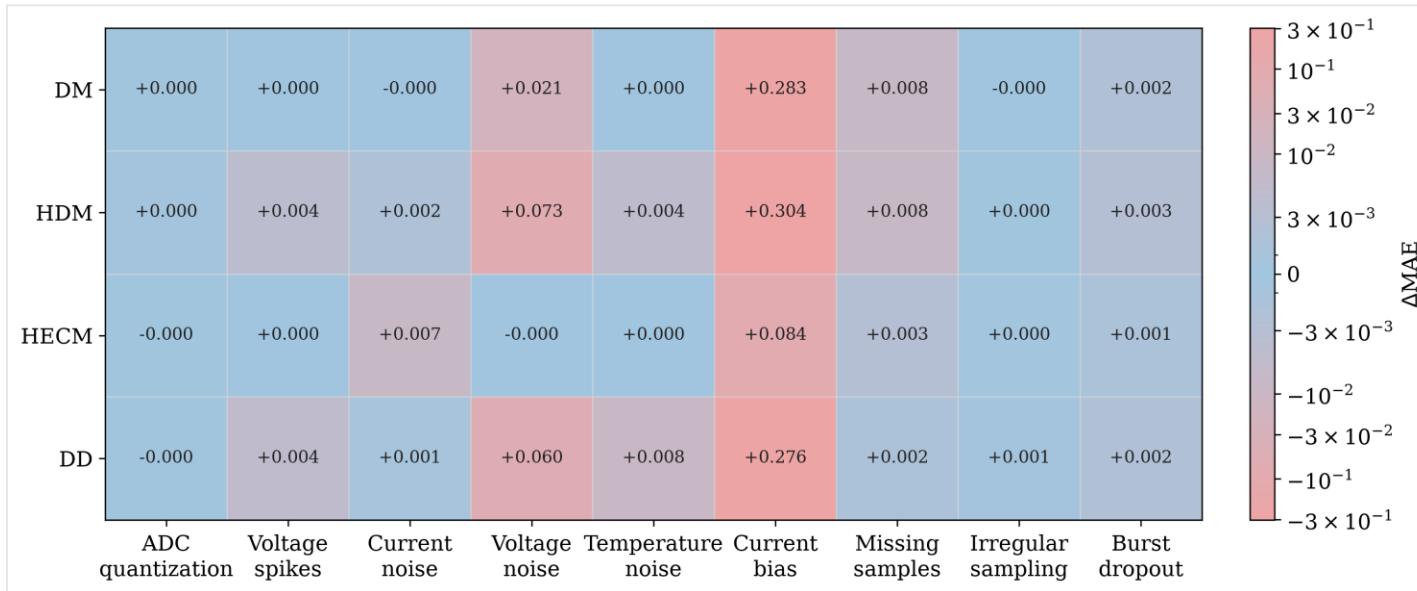
0.24 pp
HDM baseline MAE

0.90 pp
HECM baseline MAE

1.00 pp
DD baseline MAE

3.11 pp
DM baseline MAE

Disturbances Change the Estimator Ranking



- Current bias: HECM strongest
- Missing samples: DD strongest
- Initialization mismatch: DD fastest strict recovery
- Noise and spikes: estimator-specific propagation paths

Ranking shift

Fault type as decisive context

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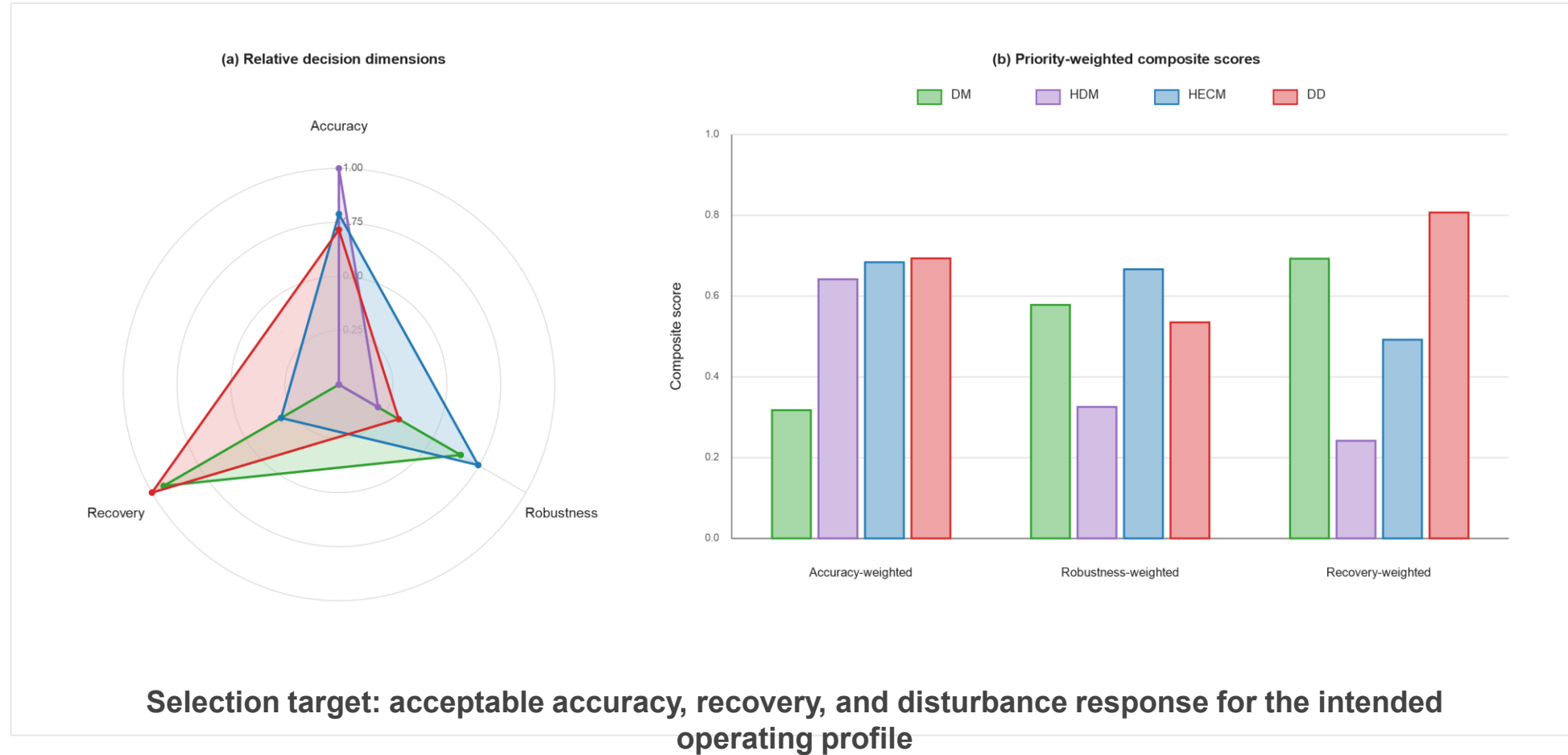
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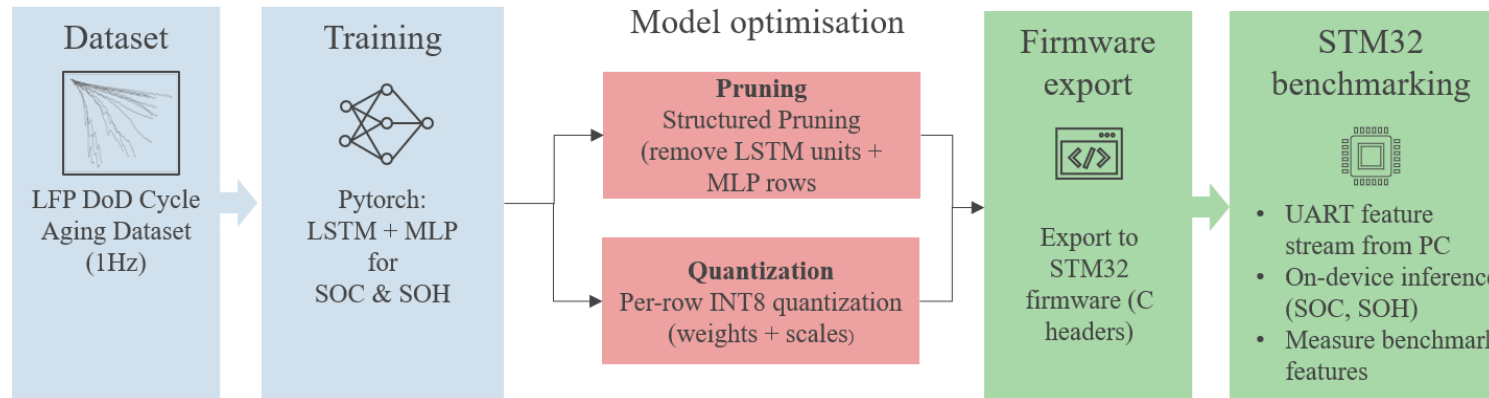
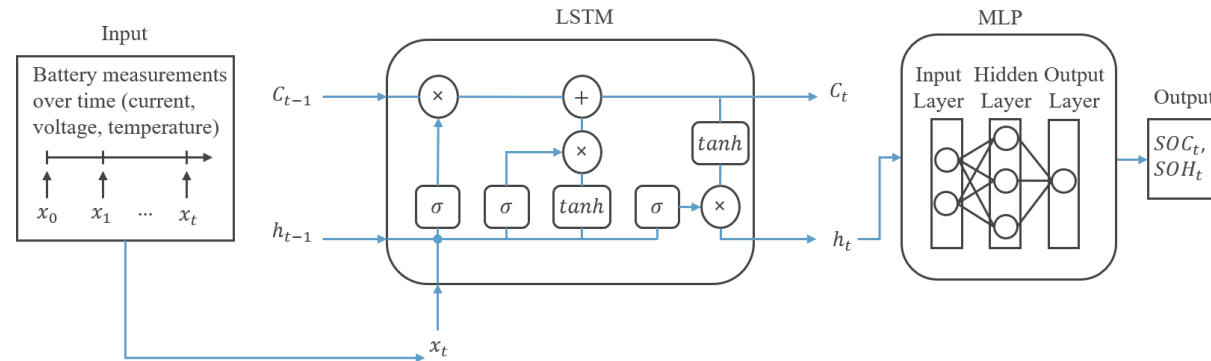
Robustness Result: No Universal Winner



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Embedded AI and Model Compression

Stateful Estimation from Training to STM32



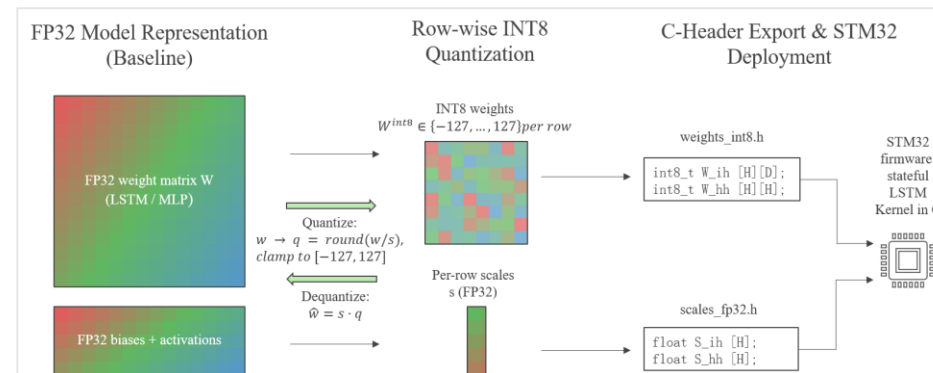
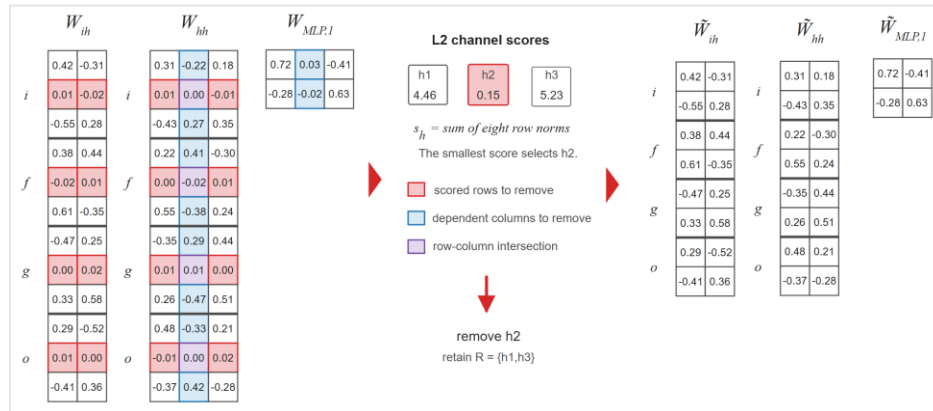
Stateful 1 Hz replay | Explicit recurrent states | PyTorch-to-C export | STM32 measurements

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Two Compression Routes

Structured pruning

INT8 weight quantization



- 30% hidden-channel reduction
- Smaller dense tensors
- Post-pruning fine-tuning

- Row-wise signed symmetric mapping
- Recurrent weights in INT8
- Scales, states, and MLP in FP32

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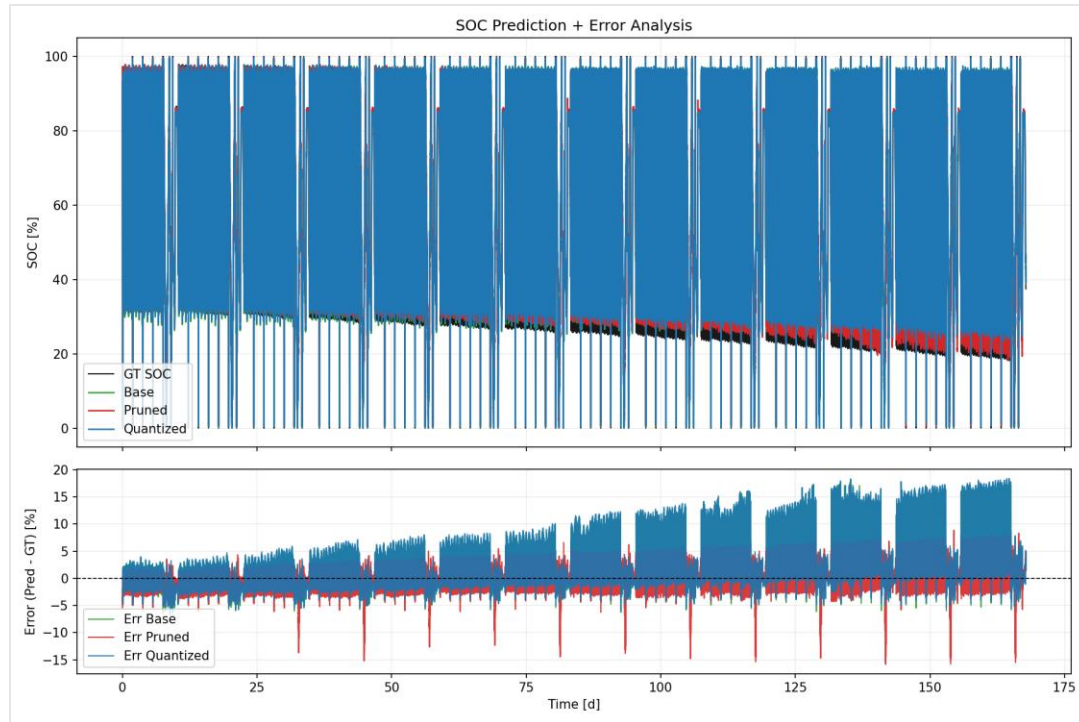
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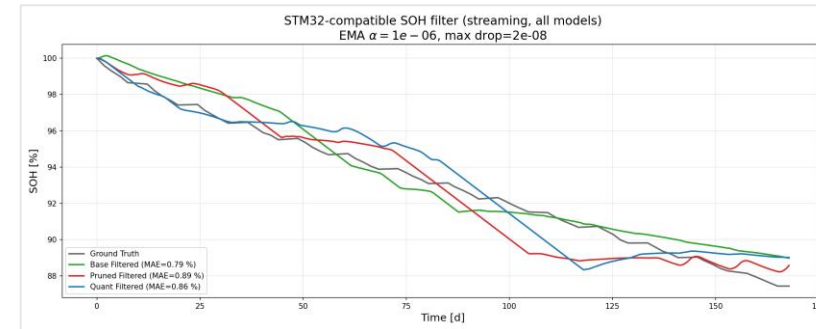
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Accuracy and Long-Horizon Stability

SOC streaming replay



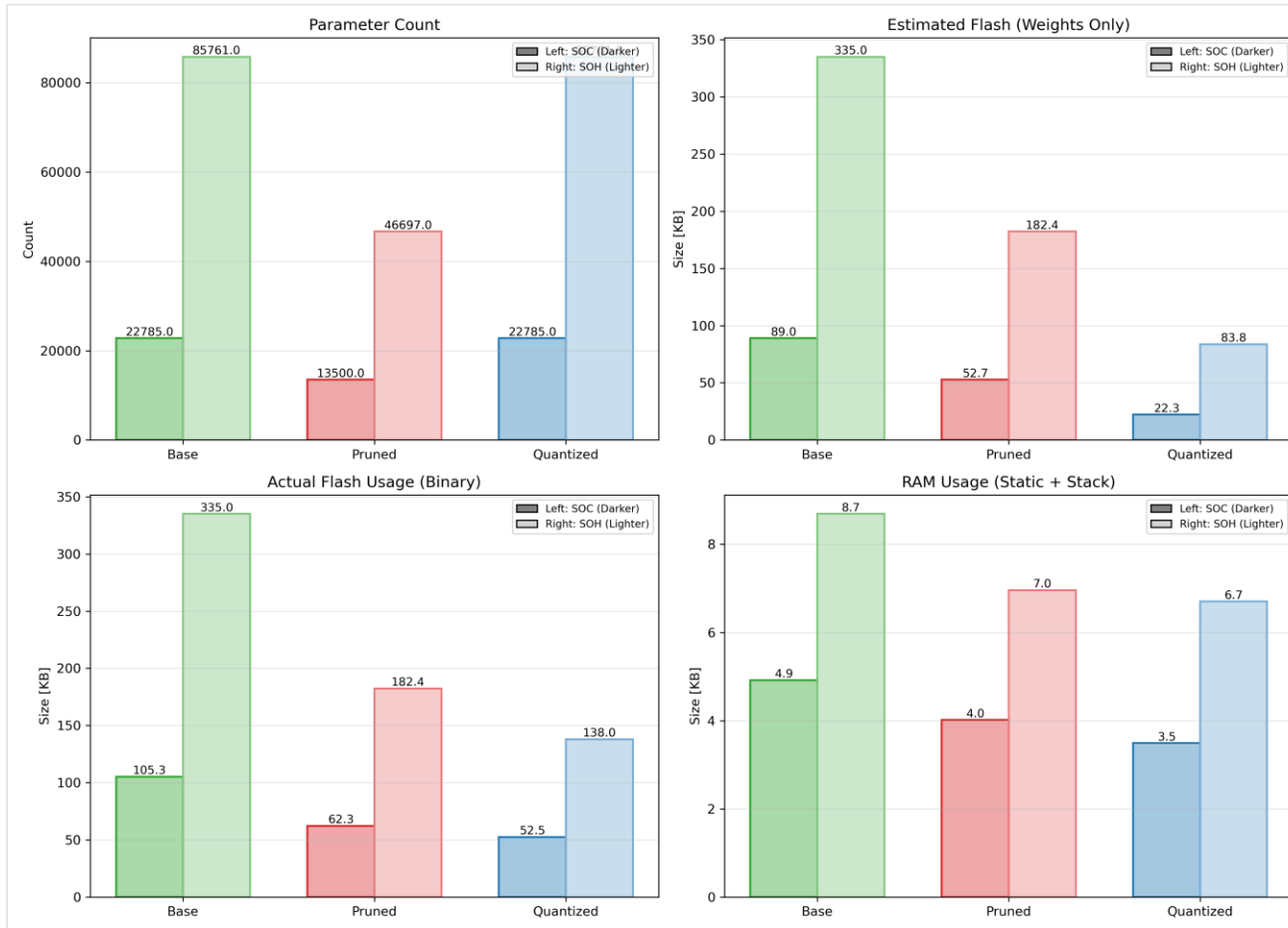
SOH streaming replay



- SOC MAE: 2.68 / 2.34 / 2.79 pp
- SOH MAE: 0.85 / 1.46 / 1.41 pp
- Order: Base / Pruned / Quantized
- No increasing Quantized-to-Base divergence

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Resource, Latency, and Utility Trade-Offs



SOC operating points

- Pruned: -41% flash, -43% kernel time
- Quantized: -50% flash, +399% kernel time
- Utility: 0.71 Pruned / 1.83 Quantized

SOH operating points

- Pruned: -46% flash, -44% kernel time
- Quantized: -59% flash, +29% kernel time
- Constraint-dependent optimum

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Integrated Findings and Outlook

Integrated Findings Across the Three Studies

1 Representation

Cumulative-current and lag features enable compact ageing estimation

2 Validation

Nominal accuracy, recovery, and disturbances require separate evidence

3 Implementation

Compression choice depends on the active hardware constraint

- Model class alone does not determine deployment suitability
- Feature pipeline, estimator, compression, firmware, and hardware as one design system
- Transferable evaluation logic instead of one universal estimator ranking

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Design Rules, Limitations, and Outlook

Design rules

- Representative features before model complexity
- Disturbance testing beyond clean MAE
- Compression matched to flash, RAM, or latency bottleneck
- Target-hardware validation before deployment

Next steps

- Additional cells, chemistries, and operating profiles
- Live disturbance injection in firmware and sensing chain
- Joint pruning and quantization
- Larger packs and longer autonomous operation

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Main conclusion

Deployment readiness as a system property

Thank You | Questions